

Parent Edu Connect - Project Documentation

AI-Based Student Monitoring & Smart Academic Analytics System

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Stack: Java Spring Boot + MySQL + Python + HTML/CSS/JS

1. Project Overview

Parent Edu Connect is a role-based academic monitoring platform for parents, teachers, and administrators. It tracks marks, attendance, behavior, achievements, and provides AI-driven predictions and a chatbot assistant.

The frontend is a responsive glass-style SPA. Data persists in MySQL via a Java REST API.

Python provides hybrid ML predictions (linear regression + rule-based scoring).

2. System Architecture

Layer 1 - Frontend: index.html, styles.css, app.js (browser)

Layer 2 - Backend: Java Spring Boot on port 8080 (REST + JPA)

Layer 3 - Database: MySQL Workbench database parent_edu_connect

Layer 4 - AI Service: Python Flask on port 5000 (predict + chat)

Flow: Browser -> Java API -> MySQL; Java API -> Python AI -> JSON response -> Browser

3. Database Schema (MySQL)

users - login accounts (parent, teacher, admin)

students - student profiles and summary metrics

marks - monthly subject marks per student

attendance - monthly attendance percentages

behavior_logs - complaints and positive remarks

achievements - certificates and awards

login_audits - login history for admin panel

active_sessions - live session tracking

school_stats - school-wide analytics counters

Scripts: database/schema.sql and database/seed.sql

MySQL password configured: 7993445360 (root user)

4. Java Backend (backend/)

ParentEduConnectApplication.java - Spring Boot entry point

entity/ - JPA entities mapping MySQL tables

repository/ - Spring Data JPA repositories

service/ - AuthService, StudentService, AiService, AdminService

controller/ApiController.java - REST endpoints under /api
config/DataSeeder.java - seeds demo data on first run
config/WebConfig.java - CORS + RestTemplate for Python calls
application.properties - MySQL JDBC URL and AI service URL

5. Python AI (python-ai/)

app.py - Flask API: /predict, /chat, /health
predictor.py - prediction formulas + sklearn LinearRegression blend
requirements.txt - flask, numpy, scikit-learn
Predictions use marks trend, attendance influence, behavior score.
Chatbot uses keyword intent matching (attendance, weak subject, predict, complaints).

6. Frontend (app.js connections)

API_BASE = http://localhost:8080/api
apiFetch() - generic HTTP helper with X-Auth-Token header
handleLogin() -> POST /api/auth/login -> loads students from MySQL
loadStudentsFromApi() -> GET /api/students
handleTeacherUpdate() -> POST /api/records (saves to MySQL)
loadPredictions() -> GET /api/predictions/{id} -> Java -> Python
addChat() -> POST /api/chat -> Python chatbot
loadAdminData() -> GET /api/admin/stats, login-audits, sessions
Falls back to local mock data if backend is offline.

7. API Endpoints

POST /api/auth/login - authenticate user
POST /api/auth/logout - end session
GET /api/students - list all students
GET /api/students/{id} - student detail
POST /api/records - teacher record entry
GET /api/predictions/{studentId}?subject=Mathematics
POST /api/chat - AI chatbot question
GET /api/admin/stats - school analytics
GET /api/admin/login-audits - login history
GET /api/admin/sessions - active sessions

8. How to Run

1. Start MySQL Workbench (root / 7993445360)

2. Run database/schema.sql and seed.sql (optional - auto-seed available)
3. cd python-ai && pip install -r requirements.txt && python app.py
4. cd backend && mvn spring-boot:run
5. Open index.html or run start-all.bat

Demo logins: parent@demo.edu/parent123, teacher@demo.edu/teacher123, admin@demo.edu/admin123

9. File Structure

index.html - main UI shell

app.js - frontend application logic + API integration

styles.css - responsive glass UI theme

backend/pom.xml - Maven dependencies

backend/src/main/java/com/parentedu/ - Java source

backend/src/main/resources/application.properties - DB config

python-ai/app.py - AI Flask server

python-ai/predictor.py - ML prediction logic

database/schema.sql - MySQL DDL

database/seed.sql - demo data

start-all.bat - Windows launcher

SETUP.md - setup instructions